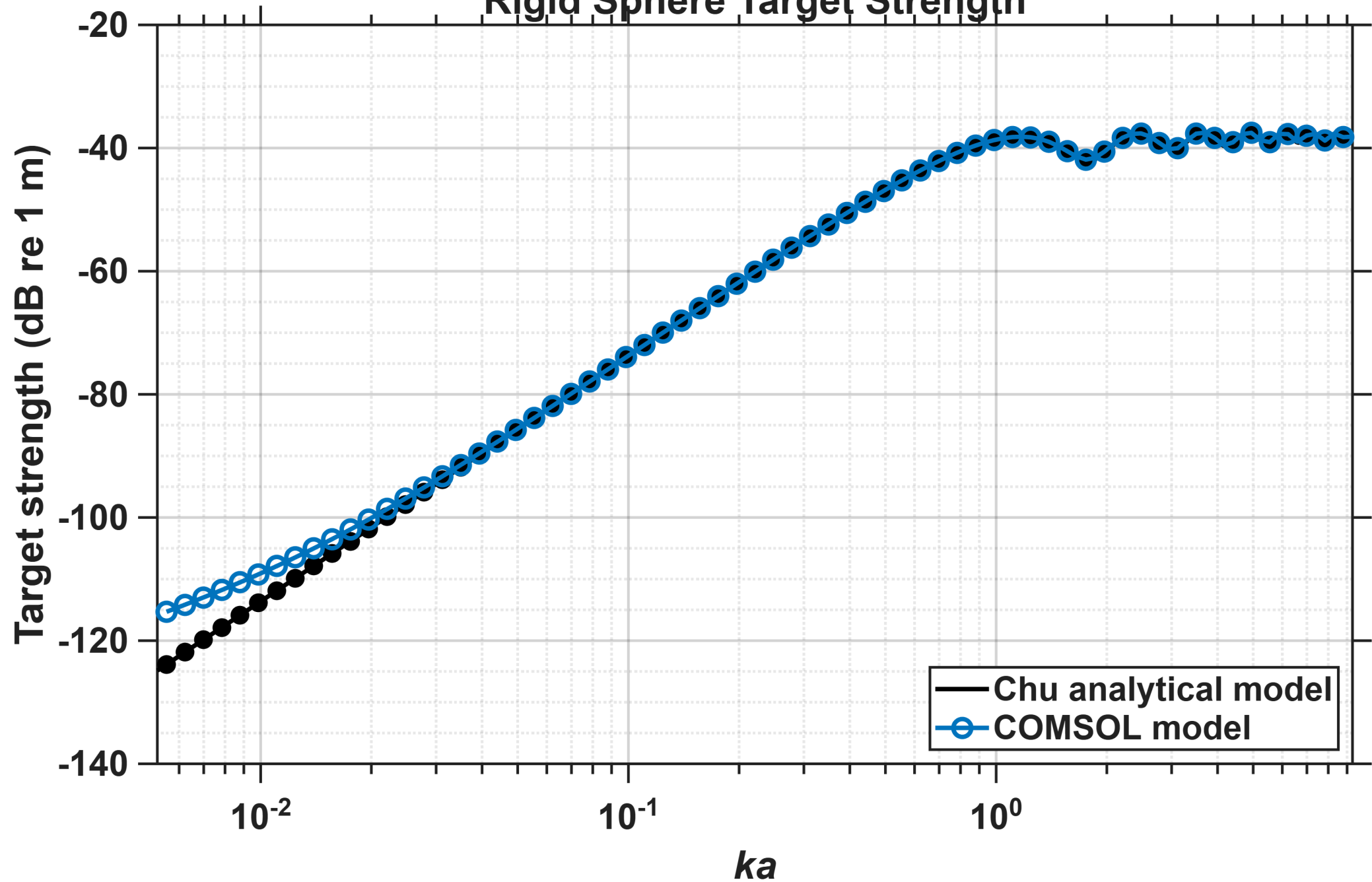
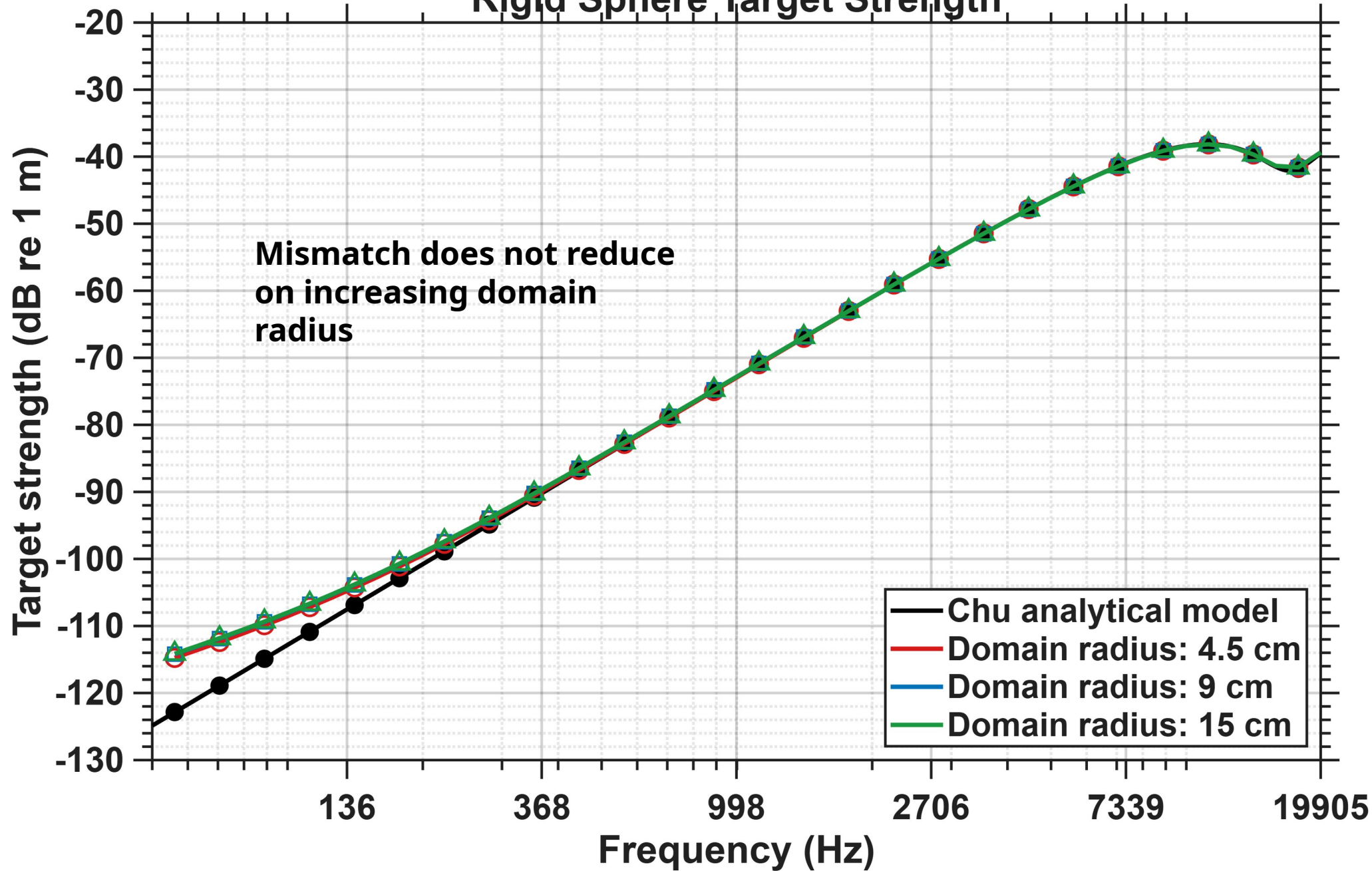


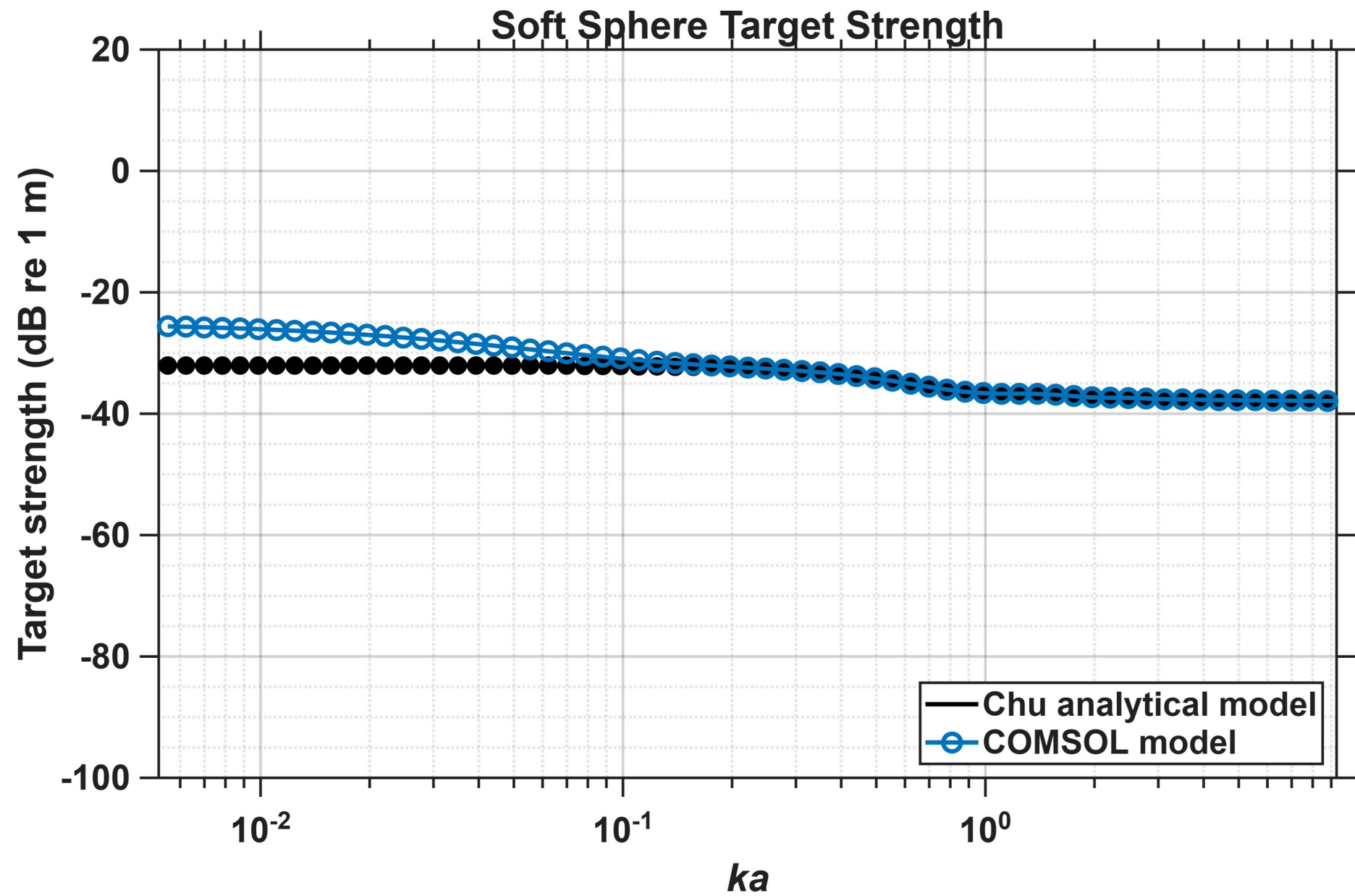
Acoustic scattering by spheres and swim bladders

Rigid Sphere Target Strength

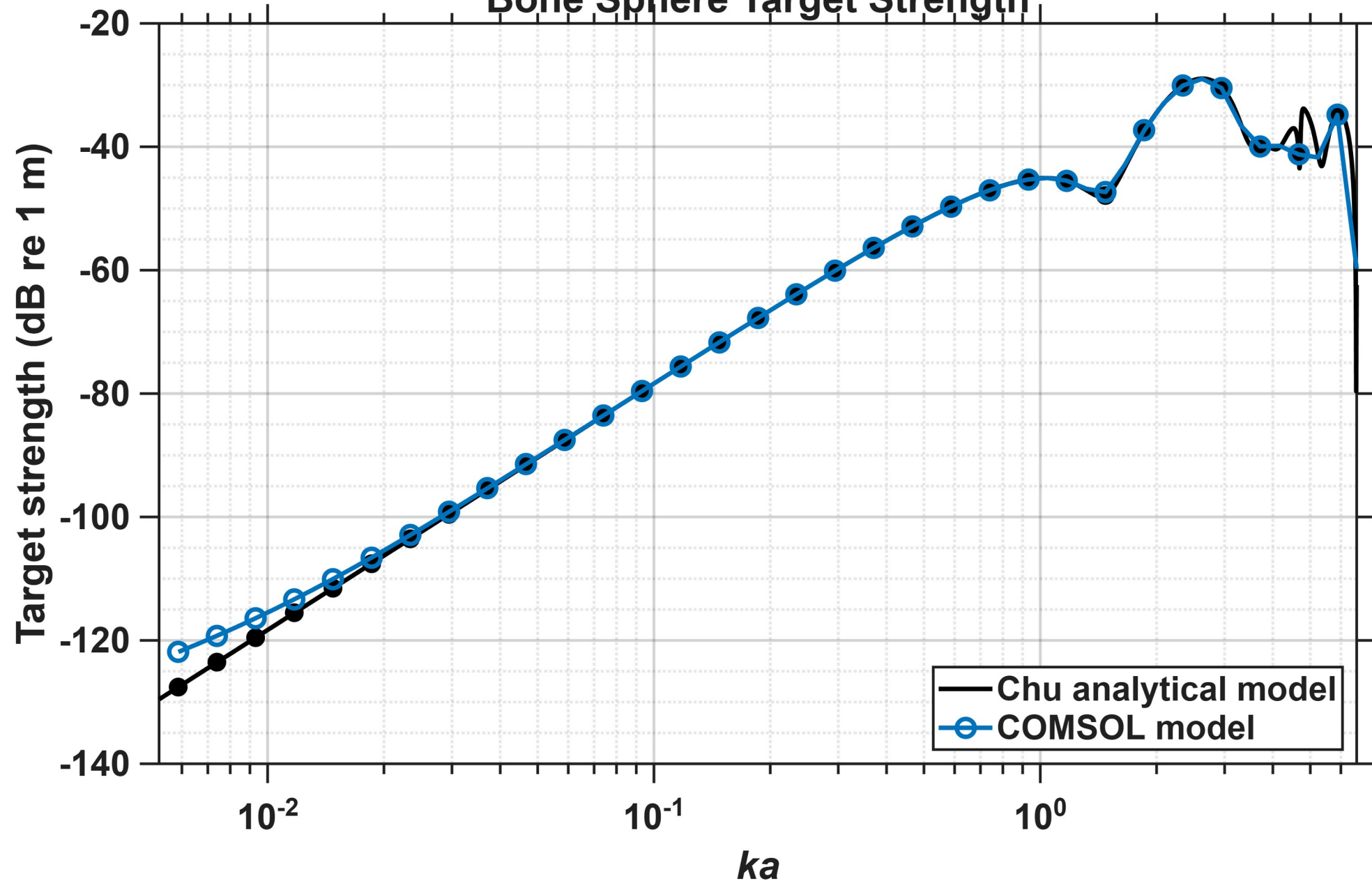


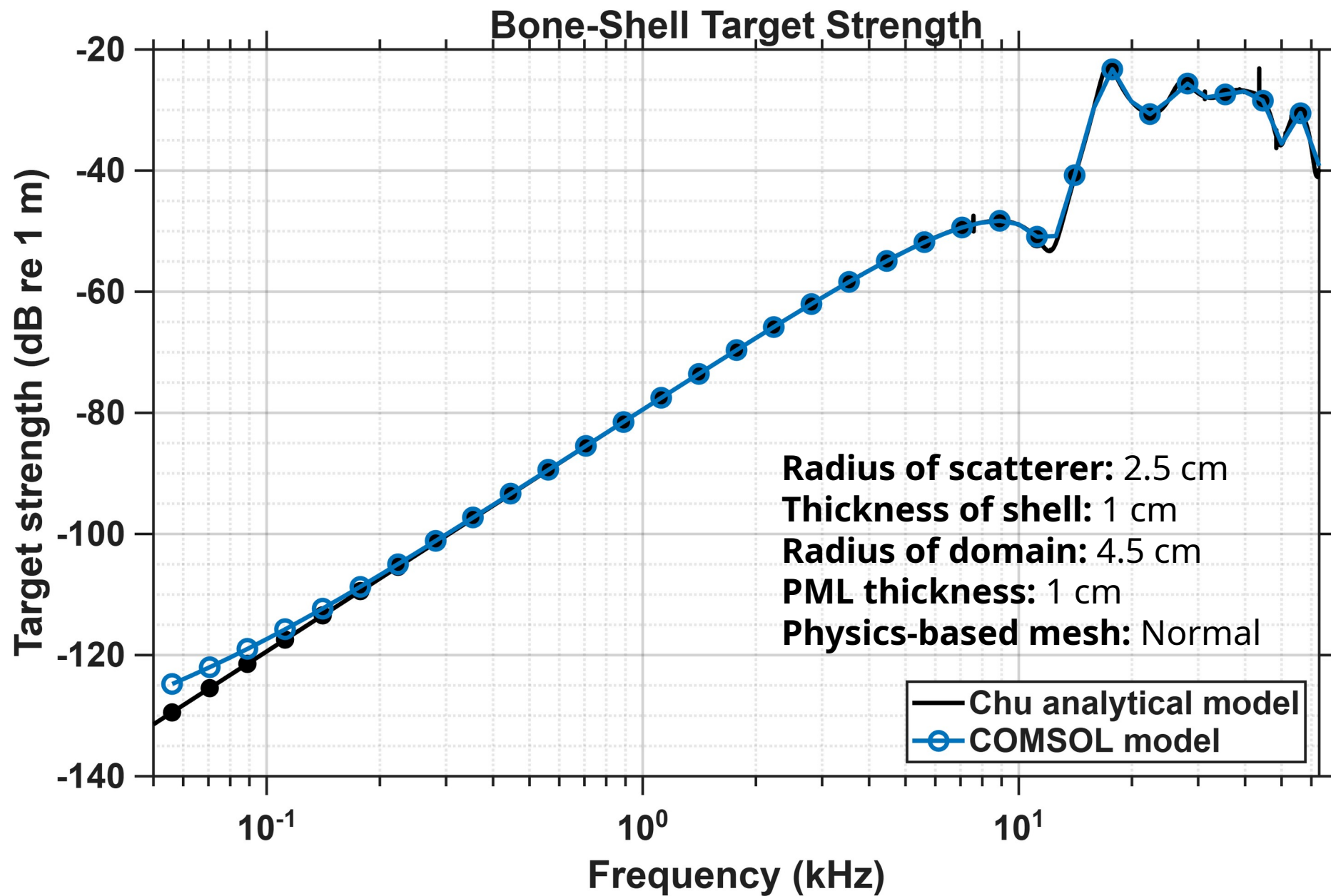
Rigid Sphere Target Strength





Bone Sphere Target Strength





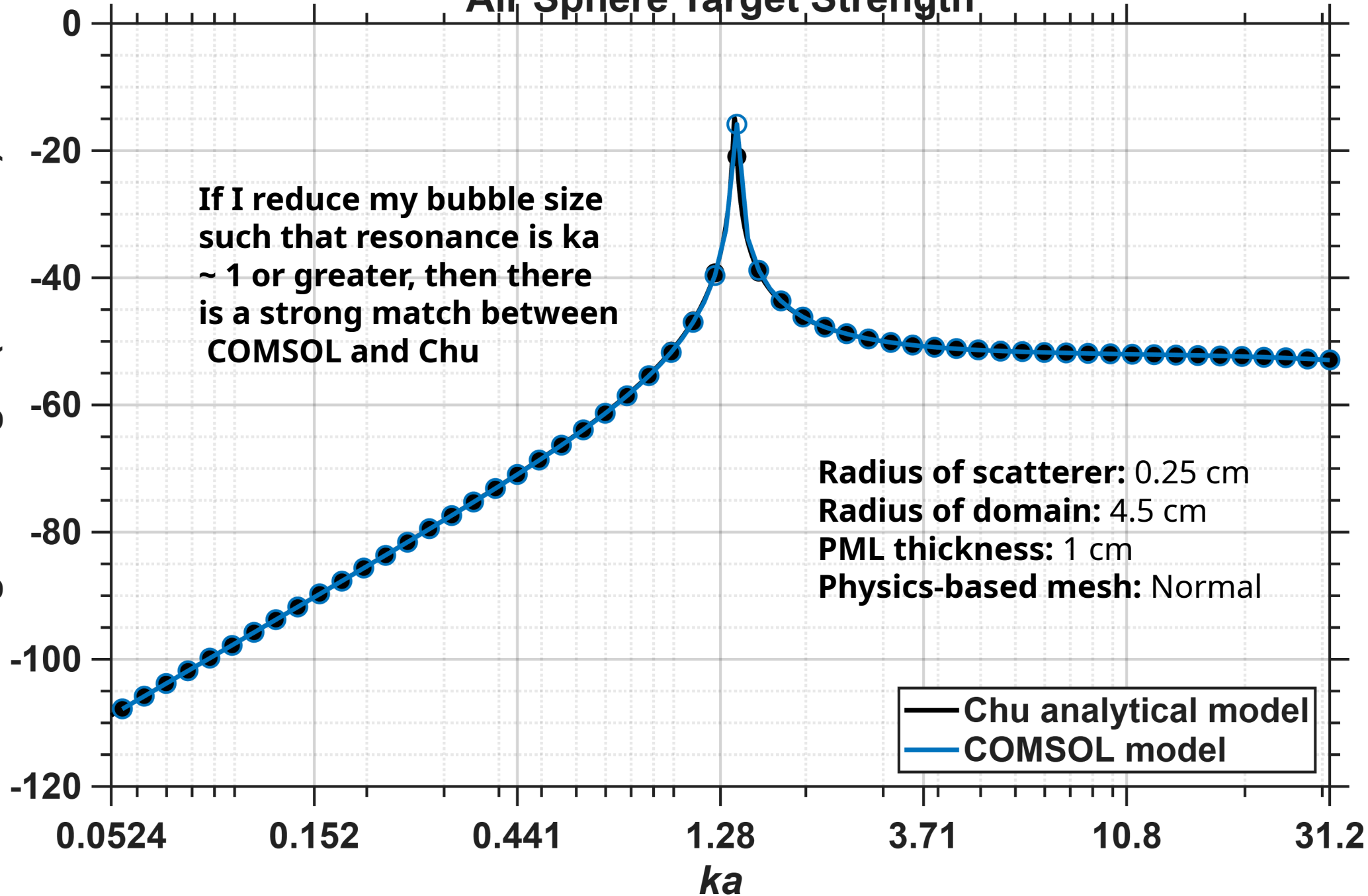
Air Sphere Target Strength

Target strength (dB re 1 m)

If I reduce my bubble size
such that resonance is ka
 ~ 1 or greater, then there
is a strong match between
COMSOL and Chu

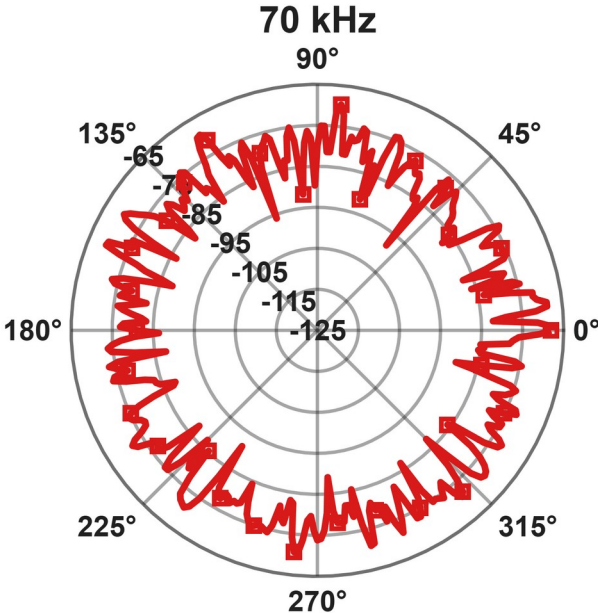
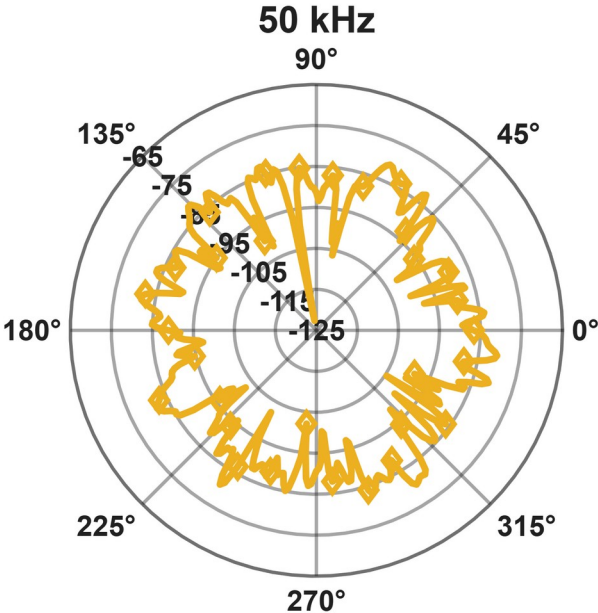
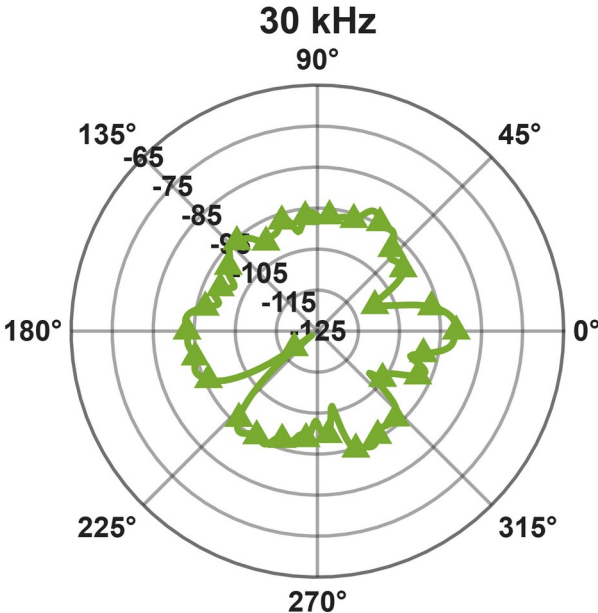
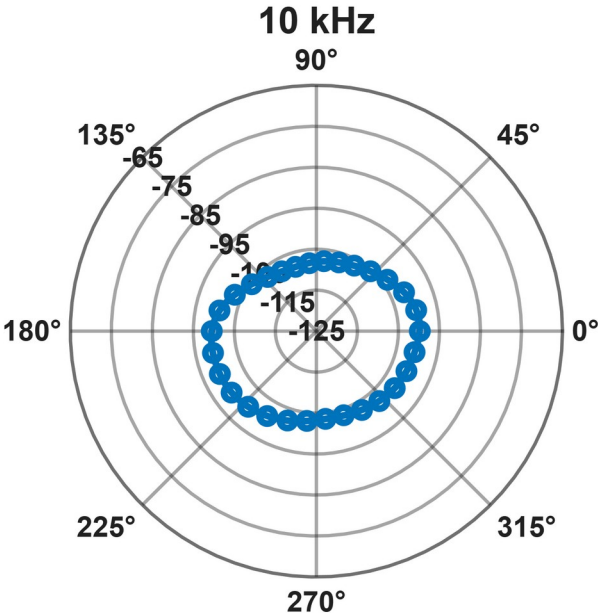
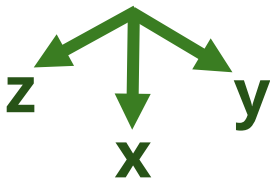
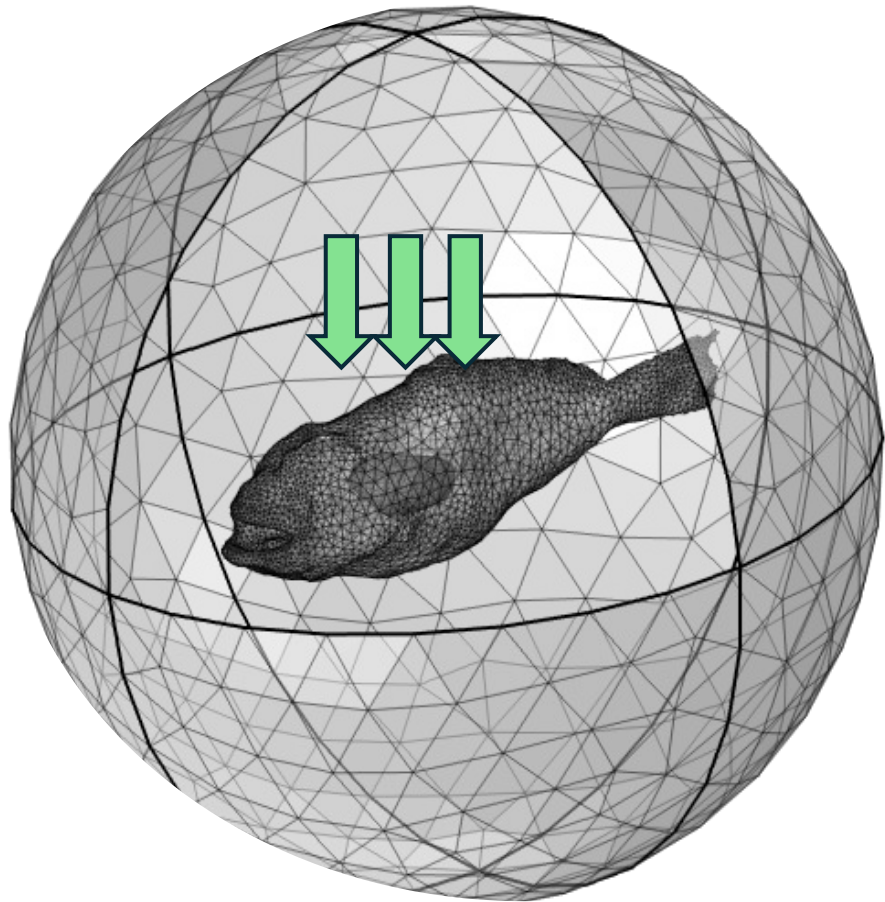
Radius of scatterer: 0.25 cm
Radius of domain: 4.5 cm
PML thickness: 1 cm
Physics-based mesh: Normal

— Chu analytical model
— COMSOL model



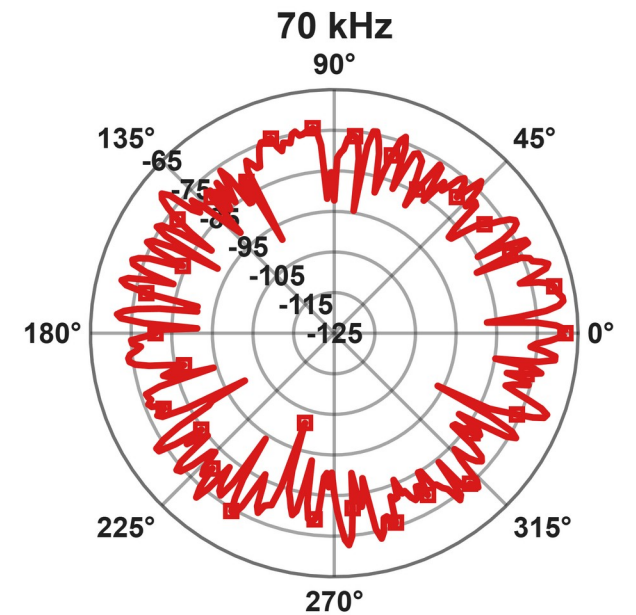
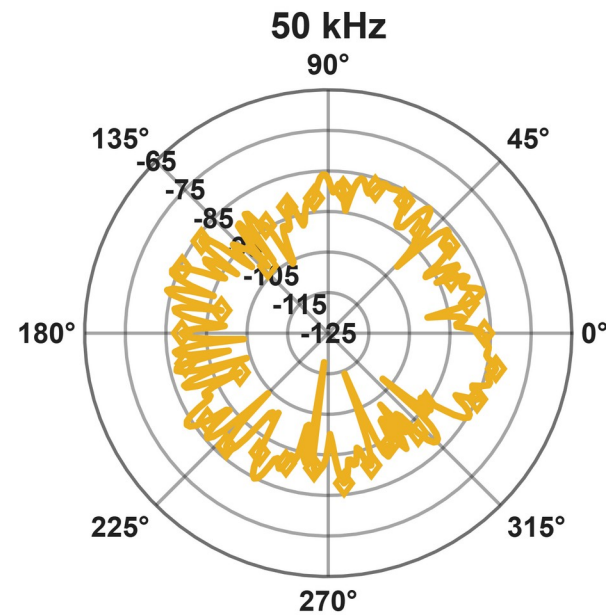
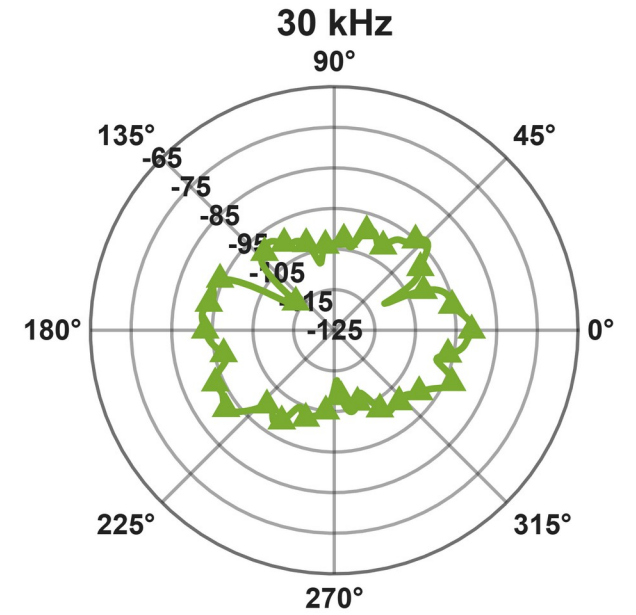
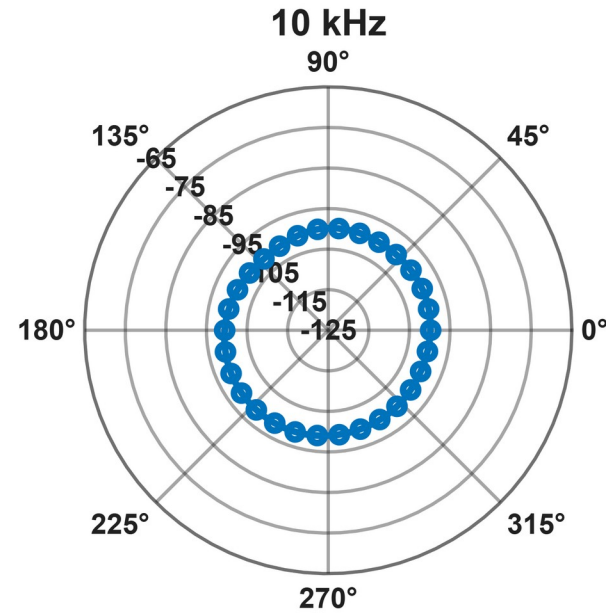
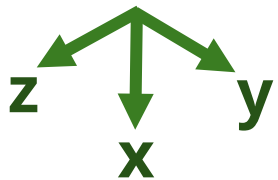
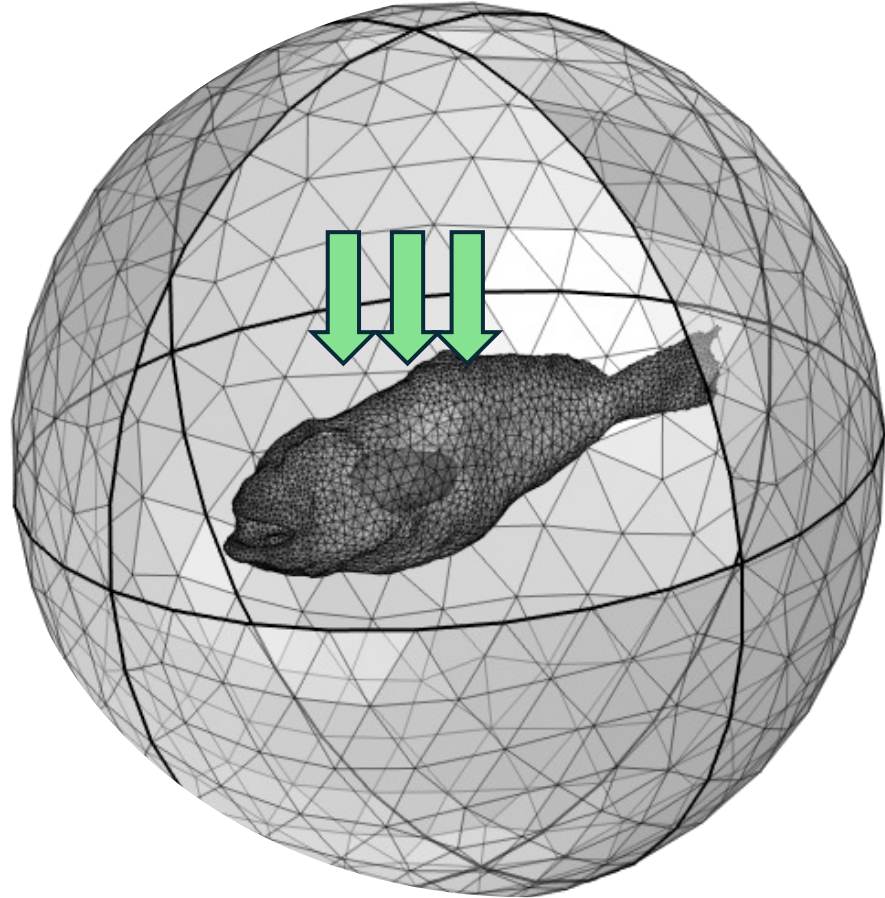
Fish and swim bladder models

Rockfish Monostatic Scattering Pattern — XZ Plane



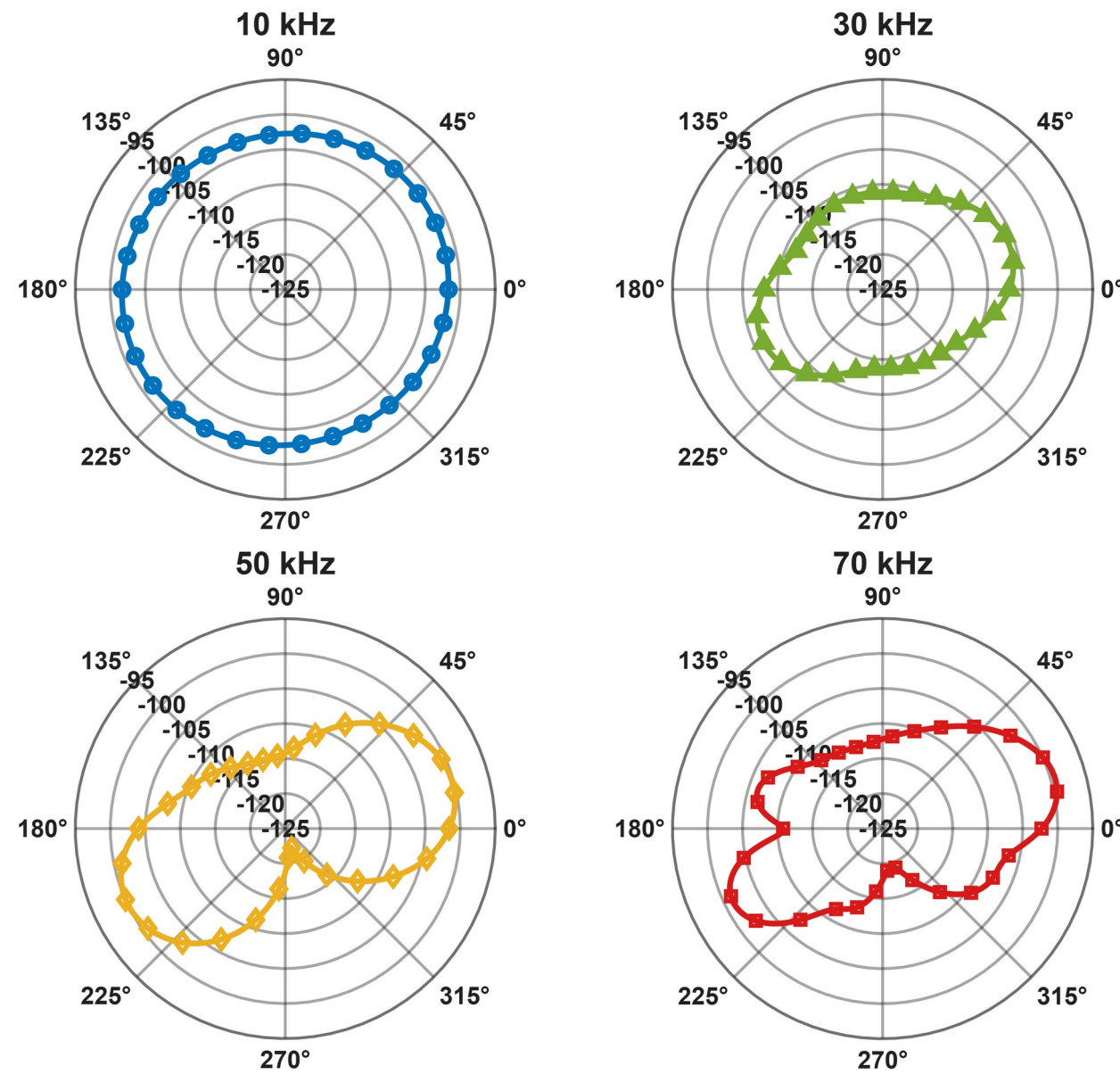
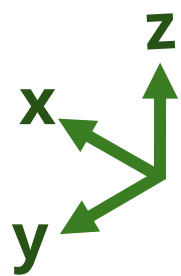
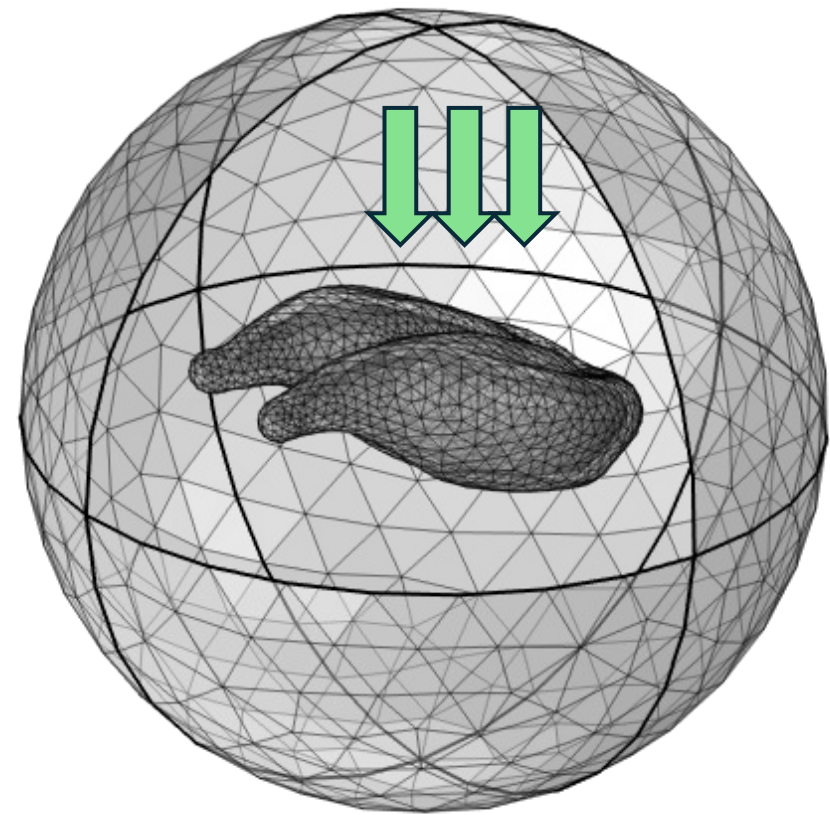
Target strength (dB re 1 m)

Rockfish Monostatic Scattering Pattern — XY Plane



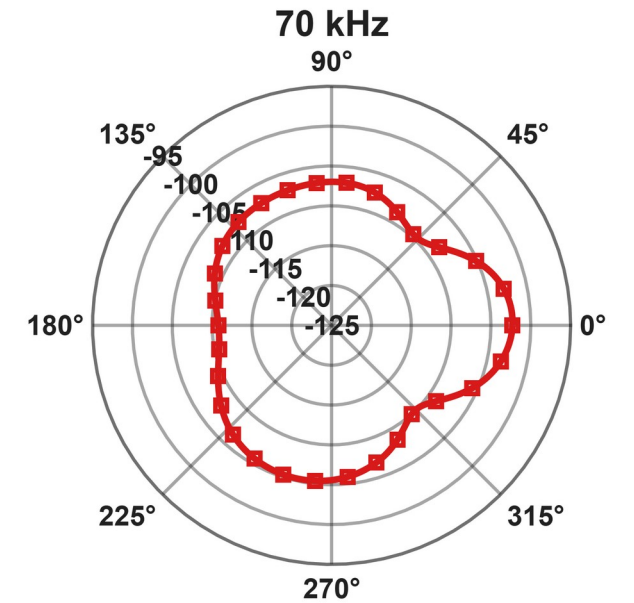
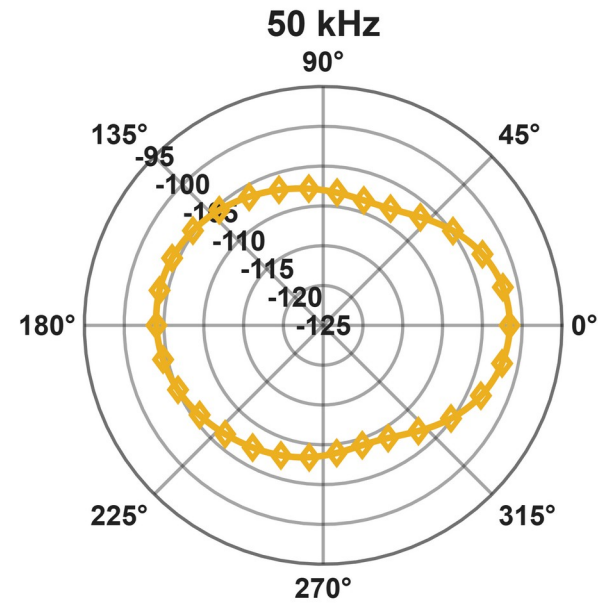
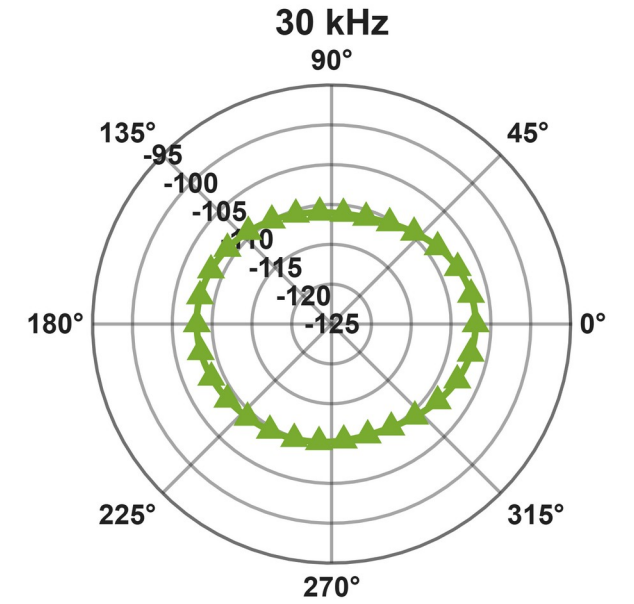
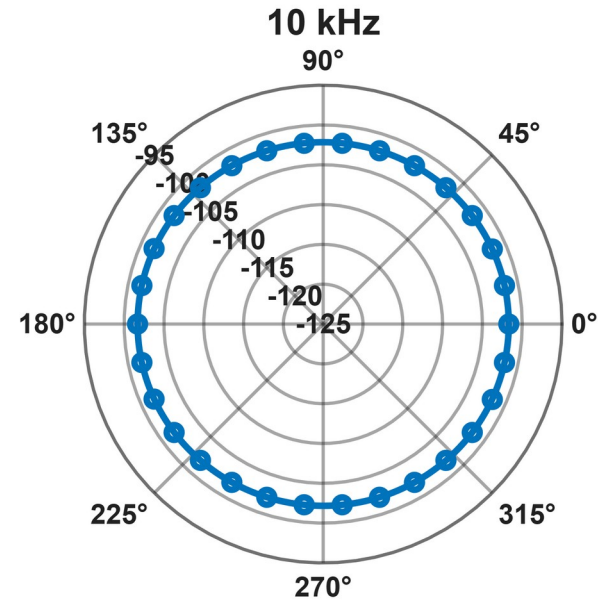
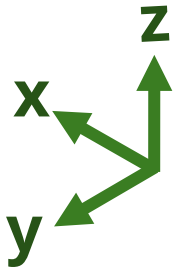
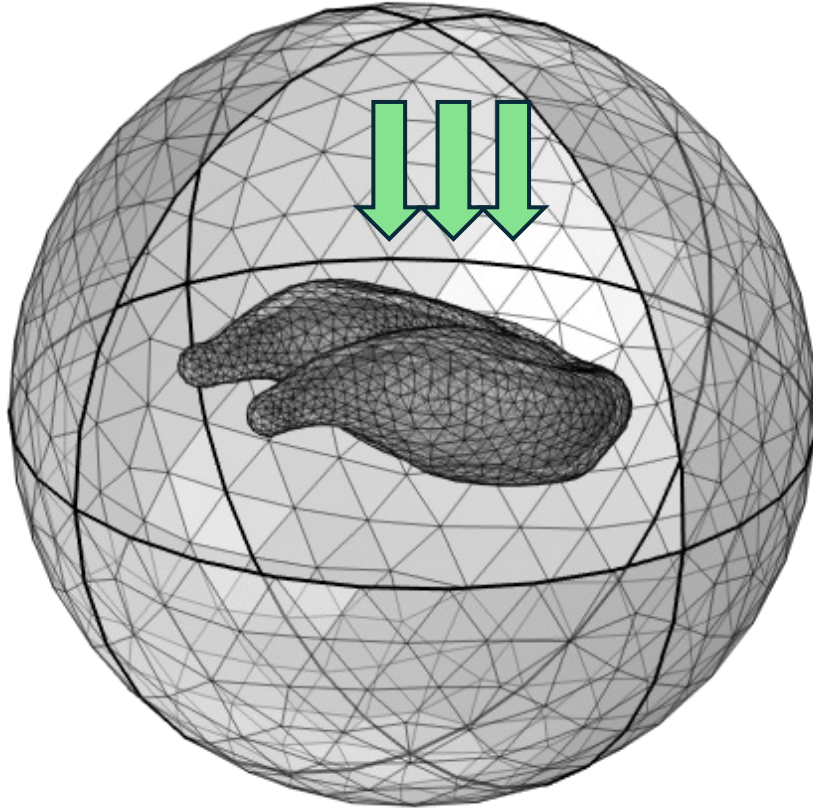
Target strength (dB re 1 m)

Midshipman Monostatic Scattering Pattern — ZY Plane



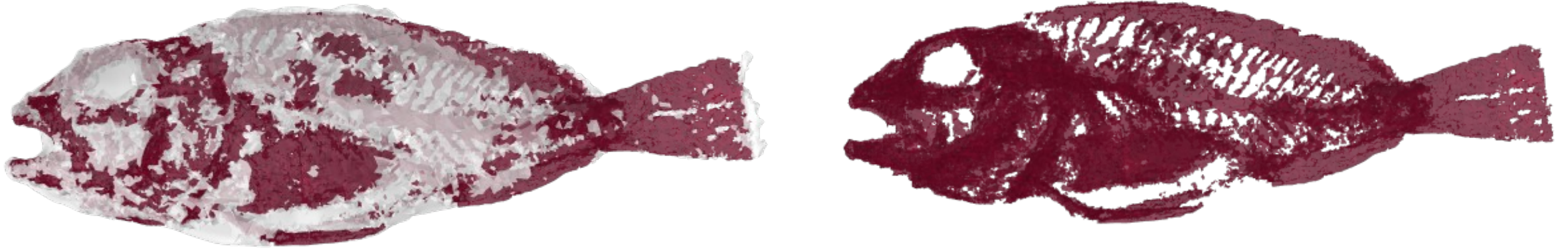
Target strength (dB re 1 m)

Midshipman Monostatic Scattering Pattern — ZX Plane



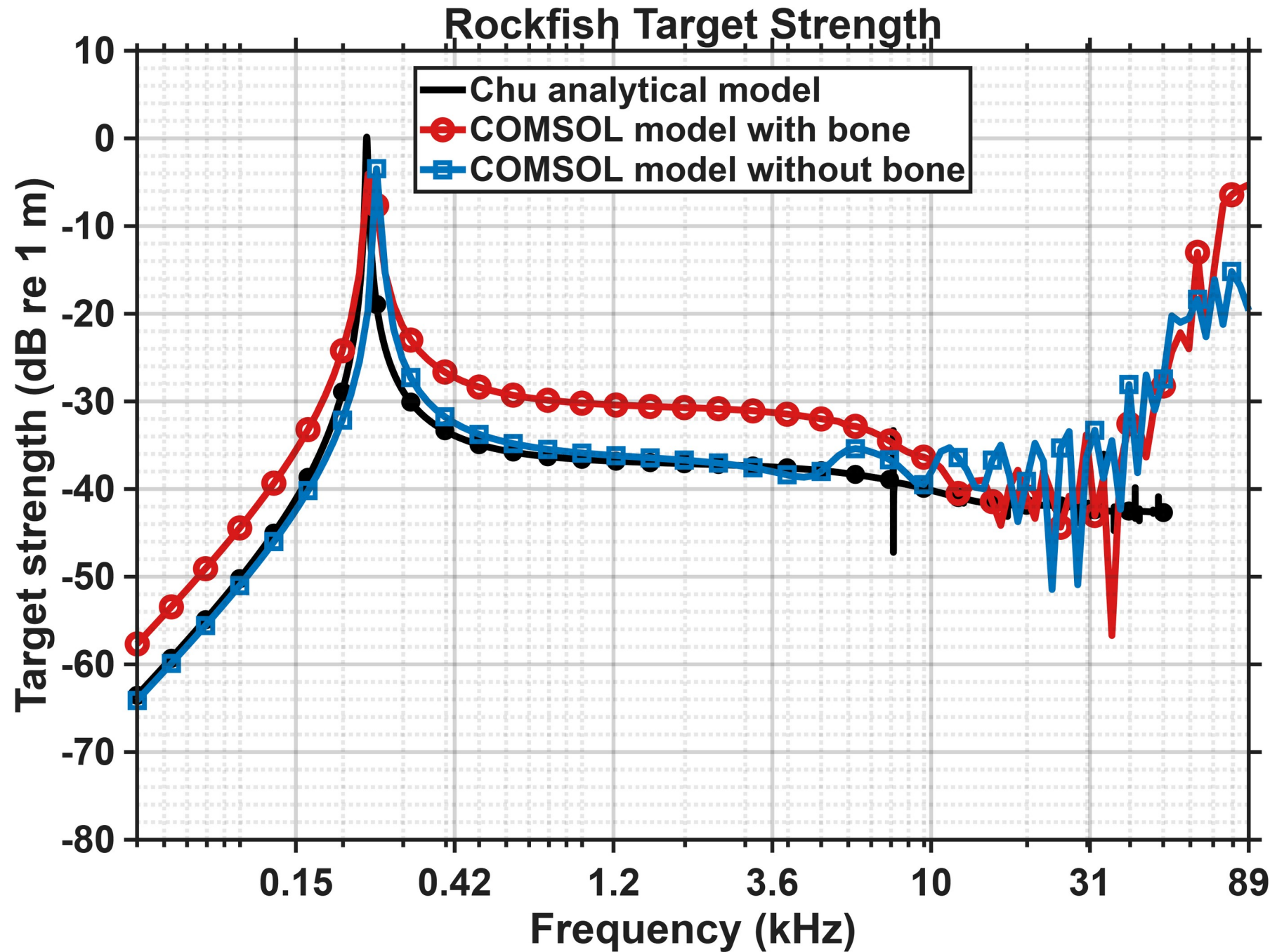
Target strength (dB re 1 m)

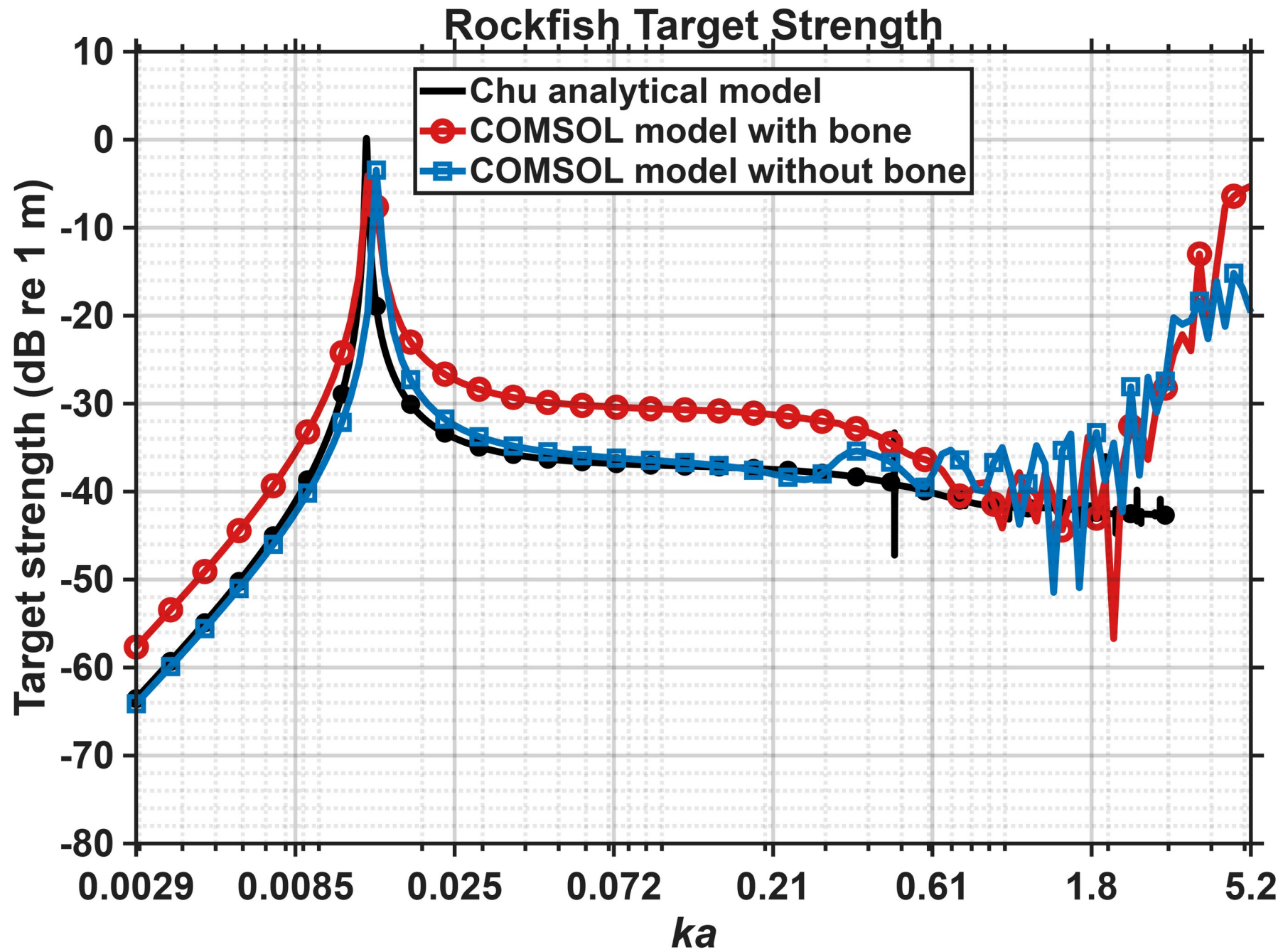
Voxel-based assignment of bone material



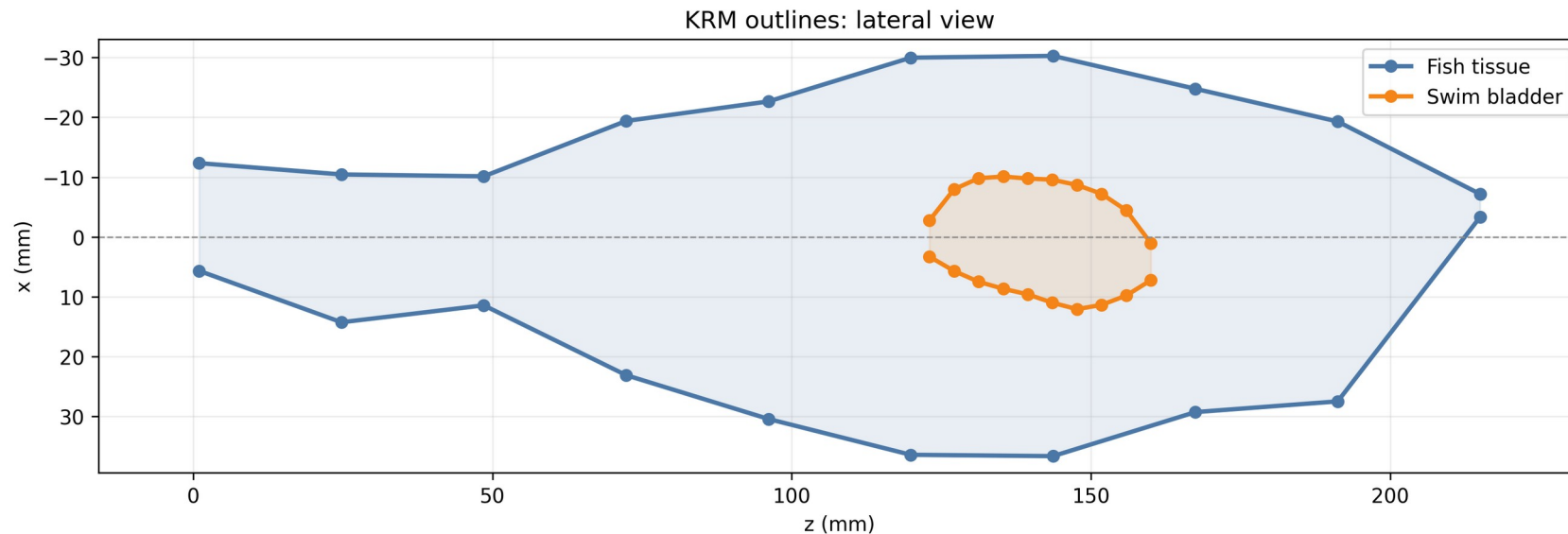
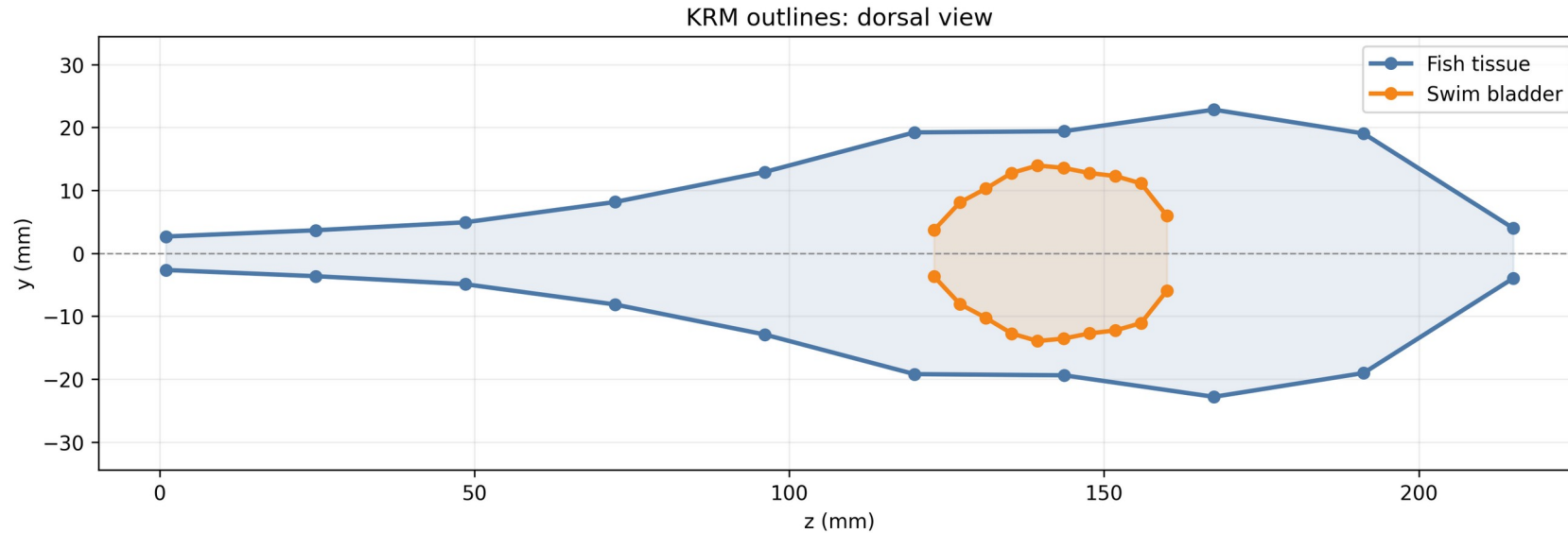
Volume plot of the binary bone mask in COMSOL (red voxels = bone; white voxels = tissue)

1. Export segmented coordinates from 3D Slicer with binary labels (1 = bone; 0 = tissue).
2. Retain only coordinates that fall within the processed fish-body surface mesh.
3. Import the data into COMSOL as a nearest-neighbor interpolation function
4. Use conditional expressions to assign bone or tissue material properties based on the interpolated label.





KRM still in progress. Here are outlines that I am using as input



For soft sphere model

Element-size distributions across frequency

